

BLOCKPAY

Cloud-enabled campus payments with blockchain-verifiable receipts

CORE IDEA

Fast off-chain payments.
Tamper-evident on-chain proof.

01

PROBLEM STATEMENT

Campus payments are convenient, but records can be disputed, altered, or trapped inside one central system.

- No independent proof of a completed transfer
- Manual bill splitting and payment follow-ups
- Centralized logs create a single point of trust

02

PROPOSED SOLUTION

A student wallet that combines cloud speed with blockchain integrity.

- Send, request, split, scan QR and notify
- Store full private records in MongoDB
- Anchor reference ID + SHA-256 hash on-chain

03

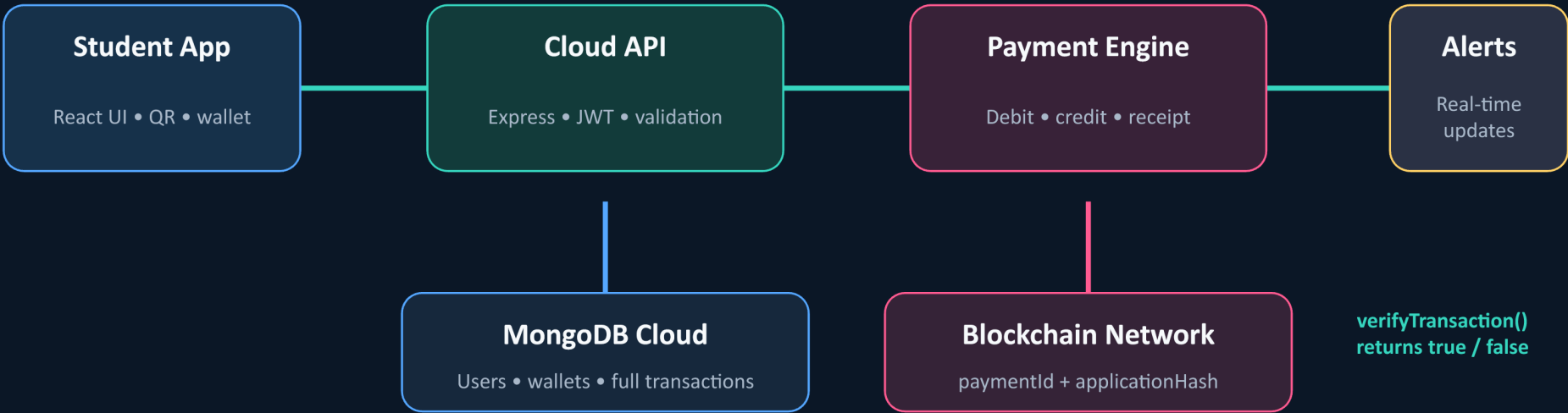
TECHNOLOGIES

React + Vite Web UI	Node + Express Cloud API
MongoDB App data	Solidity Proof registry
JWT + SHA-256 Security	Docker Deployment

04

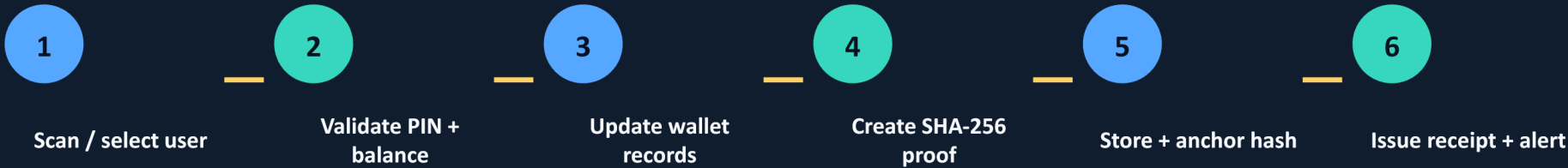
CLOUD + BLOCKCHAIN ARCHITECTURE

Operational data stays off-chain; a compact proof is mirrored on-chain for later verification.



05

PAYMENT WORKFLOW



06

EXPECTED OUTCOMES & BENEFITS

- Faster campus payments
- Auditable digital receipts
- Lower on-chain data cost
- Privacy-aware hybrid design
- Easy cloud scaling